Brian Jeong

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Education

University of Pittsburgh | *GPA: 3.7/4.0* — Pittsburgh, PA
Bachelor of Science in Computer Science & Data Science — *Expected May 2027*

Experience

Dept of Biomedical Informatics | *Clinical NLP & AI Innovation Lab (NIH Funded)* — Oct. 2026 – Present
Generative AI Software Engineer — *Pittsburgh, PA*

- Building a **multi-agent LLM framework** for Alzheimer's diagnosis on the **ADNI** dataset, orchestrating specialized agents for cognitive tests, neuroimaging, genetics, and medical history in a staged **LangGraph** workflow
- Implementing **retrieval-augmented case reasoning**, embedding historical patient cases in a **vector store** so agents can ground each diagnosis in the most similar prior cases retrieved at inference time
- Developing the **evaluation harness** that benchmarks the framework against logistic regression, random forest, and **XGBoost** baselines, controlling for **label leakage** and incorporation bias in the prior published comparison

PNC Financial Services | *Machine Learning Operations Team* — June 2026 – Aug. 2026
Software Engineering Intern — *Pittsburgh, PA*

- Migrated an unsecured-loan marketing model's raw data-science code into a production **PySpark** template, moving feature engineering and inference for 2 sub-models on a shared pipeline that scores **16.5M+ records**
- Took **sole ownership** of the model's release mid-cycle after team turnover, driving it through UAT development, QA testing, and production to meet a hard code freeze and ship the **August prod release**
- Refactored production alerting for a debit-dispute abuse model, adding a 90-day approved-dispute feature to rank the **top 2,500** flagged customers and escalate the **top 100** by financial exposure
- Designed target-state **AWS** architecture (DataSync, EMR, SageMaker) for on-prem-to-cloud batch inference
- Shipped a **Copilot Studio agent** that resolves FAQ tickets, escalates edge cases, and logs each ticket's category, volume, and resolution to SharePoint for an internal support team that fields recurring ticket types

School of Computing & Info Science | *Data Structures & Algorithms II* — Aug. 2025 – May 2026
Undergraduate Teaching Assistant — *Pittsburgh, PA*

- Instructed recitations, led office hours, and graded coursework, covering time & space complexity and optimization
- Provided implementation & student feedback, informing redesign of projects around real-world cybersecurity scenarios
- Made 20+ recitation slide decks, resulting in 15% increase in average exam scores for 70+ students across 2 semesters

Chaeum JCom | *Private Advertising Agency* — May 2025 – Aug. 2025
Design Engineering Contractor — *Seoul, South Korea*

- Built a static **Astro 5** + TypeScript portfolio for a Seoul real-estate ad agency, presenting **25 campaigns** from a typed data layer with build-time WebP image optimization and lazy loading
- Engineered an elevator-themed scroll experience in vanilla TypeScript: IntersectionObserver floor tracking, a live progress shaft, a keyboard-navigable lightbox, and reduced-motion support

Projects

CENTRAL — Church Operating System | *Next.js, Supabase, TypeScript, Anthropic API* — Aug. 2025 – Present

- Deployed a **multi-tenant** church platform used by **200+ members** of Korean Central Church of Pittsburgh, replacing FB Messenger, Google Drive, and SongSelect workflows with messaging, announcements, and planning tools
- Built an **AI slide generator** that OCRs chord charts in the browser with Tesseract.js and uses **Claude** to extract sectioned lyrics, turning a worship set into a ready-to-present, keyboard-driven lyric deck
- Modeled **7 permission tiers** and a reimbursement schema with per-fund splits and approval chains
- Load-tested chat via a custom WebSocket harness: **160 concurrent users**, **0 message loss**, 523ms p95 join
- Solo-shipped **72k+ LOC** by designing an agentic dev pipeline: parallel worktree-isolated sessions, **PreToolUse hooks** that block unscoped writes, and separation-of-duties review across specialized agents

Technical Skills

Languages: Python, Java, SQL, TypeScript, JavaScript, Go, C, R, HTML/CSS
Data & ML: PySpark, PostgreSQL, Pandas, XGBoost, LangChain, LangGraph
Cloud & Infra: AWS (EMR, SageMaker, DataSync), Docker, Kubernetes, Jenkins, Git, Linux
Web: React, Next.js, Django, Supabase, TailwindCSS, Vercel